

# FIBER–GRAND: Revised Claim Contract and Decisive Strong-Baseline Gate

Certified aggregate inverse search for bounded synchronization errors

Scientific de-risking dossier, version 2.0

6 August 2026

## Scientific status

The first kill-fast gate established a correct and substantial finite-block computational mechanism, but it also invalidated the proposal’s claim that “fiber guesswork” is a new information-theoretic random variable. The revised programme is authorized for exactly one further decisive gate. It must survive (i) codebook-free execution, (ii) an independent code-specific exact baseline, (iii) a new fixed-edit complexity theorem, and (iv) specialized-code comparison. A positive outcome still does not establish field-defining status; novelty adjudication and a calibrated physical impairment remain mandatory.

## Contents

|          |                                                |          |
|----------|------------------------------------------------|----------|
| <b>1</b> | <b>Purpose and decision discipline</b>         | <b>2</b> |
| 1.1      | The three possible outcomes . . . . .          | 2        |
| <b>2</b> | <b>Revised novelty boundary</b>                | <b>2</b> |
| 2.1      | Claims that are withdrawn . . . . .            | 2        |
| 2.2      | Surviving claim . . . . .                      | 3        |
| <b>3</b> | <b>Channel and exact aggregate decoder</b>     | <b>3</b> |
| <b>4</b> | <b>New fixed-edit shell theorem</b>            | <b>4</b> |
| <b>5</b> | <b>Independent strong exact baselines</b>      | <b>5</b> |
| 5.1      | Syndrome-trellis aggregate search . . . . .    | 5        |
| 5.2      | Code-aware aggregate prefix $A^*$ . . . . .    | 5        |
| 5.3      | Specialized VT control . . . . .               | 5        |
| <b>6</b> | <b>Frozen computational protocol</b>           | <b>5</b> |
| <b>7</b> | <b>Decision contract</b>                       | <b>5</b> |
| 7.1      | Hard failures . . . . .                        | 5        |
| 7.2      | Continuation threshold . . . . .               | 6        |
| <b>8</b> | <b>Reference validation and interpretation</b> | <b>6</b> |

|           |                                                                 |          |
|-----------|-----------------------------------------------------------------|----------|
| <b>9</b>  | <b>Final field-defining gate after a positive version-2 run</b> | <b>7</b> |
| 9.1       | N1: independent primary-source novelty adjudication . . . . .   | 7        |
| 9.2       | N2: independent proof review . . . . .                          | 7        |
| 9.3       | N3: additional specialized system baselines . . . . .           | 7        |
| 9.4       | N4: calibrated post-front-end impairment . . . . .              | 7        |
| 9.5       | Final stop rule . . . . .                                       | 8        |
| <b>10</b> | <b>Use-case boundary</b>                                        | <b>8</b> |
| <b>11</b> | <b>Reproducibility requirements</b>                             | <b>8</b> |
| <b>12</b> | <b>Primary-source boundary</b>                                  | <b>8</b> |
| <b>13</b> | <b>Conclusion</b>                                               | <b>9</b> |

# 1 Purpose and decision discipline

The original programme proposed exact maximum-likelihood (ML) decoding for non-injective random-transformation channels by summing all latent histories that support each candidate input. The first complete channel is exactly one uniformly located deletion combined with independent binary substitutions. The decisive observation is valid:

$$W(y | x) = \sum_{z: T_z(x)=y} p(z | x),$$

so a best individual history need not identify the best aggregate candidate. The first gate certified the likelihood formula, runtime certificate, and multiple code interfaces, and found large gains relative to exhaustive codeword-wise ML. It also exposed the limits of that evidence:

- (a) exhaustive codeword ML is not the strongest code-specific exact baseline;
- (b) the implementation used pre-enumerated codeword tables for membership at the largest lengths;
- (c) fixed-edit breadth was supported by very small samples; and
- (d) aggregate posterior rank is ordinary conditional guesswork under a uniform prior.

This dossier freezes the revised claim before further computation. It prohibits silently returning to the original terminology if the revised gate fails.

## 1.1 The three possible outcomes

### **CONTINUE\_REVISIED\_FIBER\_TRACK\_TO\_FINAL\_GATE**

The fixed-edit theorem, codebook-free decoder, and strong exact baselines pass. Only an independent novelty/proof audit, specialized system baselines, and a calibrated physical impairment are authorized next.

### **NARROW\_TO\_FIXED\_EDIT\_FOUNDATIONS\_STOP\_FIELD\_DEFINING**

A theorem or useful exact algorithm survives, but the membership-only decoder is not competitive with code-specific methods. A bounded-edit foundations paper may proceed; the multi-year field-defining programme stops.

### **STOP\_FIBER\_FIELD\_DEFINING\_PROGRAMME**

Correctness, the fixed-edit theorem, or the strong-baseline mechanism fails. No deeper simulation, hardware, or general-transducer work is permitted.

# 2 Revised novelty boundary

## 2.1 Claims that are withdrawn

The following claims are not part of the revised programme.

1. *Fiber guesswork is a new information-theoretic object.* For a uniform prior over the candidate universe, ranking  $x$  by  $W(y | x)$  is exactly optimal posterior guessing of  $X$  given  $Y$ . Conditional guessing and its Rényi bounds are classical.
2. *Priority queues, admissible bounds, or path summation are themselves new.* Priority-first ML decoding, sequential synchronization decoding, HMM/weighted-transducer path summation, and best-string versus best-path distinctions are established.

3. *The current experiments prove superiority over synchronization coding.* They compare mainly with exhaustive codeword likelihood and therefore establish a mechanism, not state-of-the-art dominance.
4. *Unrestricted weighted transducers admit efficient exact code-agnostic decoding.* Most-probable-string problems in ambiguous probabilistic automata can be NP-hard; unrestricted efficiency is not assumed.

## 2.2 Surviving claim

The only candidate field-defining residue is the following synthesis:

For bounded synchronization edits, exact aggregate inverse search can retain the code-modular membership interface of GRAND, provide an auditable ML certificate, and incur only a controlled fixed-edit overhead above the substitution-noise guessing exponent. The programme seeks a theorem characterizing this tractable region and a decoder that remains competitive with code-specific exact search.

The novelty must ultimately reside in a theorem-and-algorithm package, not terminology. The primary-source audit must separately compare the result with GRAND, priority-first block decoding, Gallager-style synchronization sequential decoding, VT/GC+/marker/watermark methods, weighted automata, and the most-probable-string literature.

## 3 Channel and exact aggregate decoder

Let  $x \in \{0,1\}^n$ . Exactly  $t$  positions are selected uniformly without replacement and deleted. The  $m = n - t$  survivors pass through a BSC with crossover probability  $p < 1/2$ . The received word is  $y \in \{0,1\}^m$ . Let

$$\mathcal{J}_t = \{J \subseteq [n] : |J| = t\}, \quad S = |\mathcal{J}_t| = \binom{n}{t}.$$

For  $J \in \mathcal{J}_t$ , let  $d_J(x, y)$  be the Hamming mismatch between  $y$  and the length- $m$  subsequence of  $x$  obtained by deleting  $J$ . The exact candidate likelihood is

$$\Lambda_y(x) = \frac{1}{S} \sum_{J \in \mathcal{J}_t} p^{d_J(x,y)} (1-p)^{m-d_J(x,y)}. \quad (1)$$

For each  $J$ , an inverse stream enumerates a survivor-error mask  $e \in \{0,1\}^m$  and  $t$  hidden deleted bits. Histories are emitted by increasing  $w_H(e)$ . The component probability in each stream is

$$\omega_w = \frac{1}{S} p^w (1-p)^{m-w}. \quad (2)$$

The same  $x$  can arise in several streams, so the decoder canonicalizes candidates. At first discovery it calls a code-membership oracle. Only codewords are completely scored by (1); noncodewords cannot compete in the code-constrained ML maximization.

If  $\eta_J$  is the next unprocessed component probability in stream  $J$ , every completely unseen candidate obeys

$$\Lambda_y(x) \leq U_{\text{new}} := \sum_{J \in \mathcal{J}_t} \eta_J. \quad (3)$$

Once the best discovered codeword score  $s^*$  satisfies  $s^* > U_{\text{new}}$ , it is the exact ML decision. A complete tie set requires retaining all discovered codewords at  $s^*$  and strict exclusion of unseen ties.

## 4 New fixed-edit shell theorem

This section states the main theorem whose independent proof review is required. It is more informative than the elementary posterior-rank identity because it bounds the work of a specific code-modular certified algorithm.

**Theorem 1** (Deterministic shell-certificate bound). *Consider the exactly- $t$ -deletion/BSC channel above with  $0 < p < 1/2$ . Let  $E$  be the actual number of substitutions on the surviving symbols in the transmitted history and set*

$$a = \frac{1-p}{p}, \quad L_{n,t} = \left\lfloor \log_a \binom{n}{t} \right\rfloor, \quad w_E = \min\{m, E + L_{n,t}\}.$$

*Run the history enumerator through every stream component of substitution weight at most  $w_E$ . By that point, the exact residual-mass certificate must have fired. Consequently,*

$$Q_{\text{hist}} \leq 2^t \binom{n}{t} \sum_{i=0}^{w_E} \binom{m}{i}. \quad (4)$$

*For  $p = 0$ , the corresponding bound is  $Q_{\text{hist}} \leq 2^t \binom{n}{t}$ .*

*Proof.* The true codeword is discovered no later than the actual deletion subset, actual hidden bits, and actual survivor-error mask, which lies in shell  $E$ . Its aggregate likelihood is therefore at least the probability of this one supporting history:

$$s^* \geq \frac{1}{S} p^E (1-p)^{m-E}.$$

After every component of weight at most  $w$  has been emitted, the next component in each of the  $S$  streams has weight at most  $S^{-1} p^{w+1} (1-p)^{m-w-1}$ . Hence (3) gives

$$U_{\text{new}} \leq p^{w+1} (1-p)^{m-w-1}.$$

At  $w = E + L_{n,t} < m$ ,

$$\frac{U_{\text{new}}}{S^{-1} p^E (1-p)^{m-E}} \leq S \left( \frac{p}{1-p} \right)^{L_{n,t}+1} < 1,$$

because  $L_{n,t} + 1 > \log_a S$ . Thus the strict certificate holds. If  $w_E = m$ , all streams are exhausted and the residual mass is zero. Each stream contains  $2^t \binom{m}{i}$  histories in shell  $i$ , yielding (4). At  $p = 0$ , only shell zero has positive mass.  $\square$

**Corollary 1** (Fixed edits preserve the BSC history-work exponent). *Fix  $t$  and  $p \in (0, 1/2)$ . For every  $\epsilon > 0$ ,*

$$\Pr \left\{ \frac{1}{n} \log_2 Q_{\text{hist}} \leq h_2(p) + \epsilon \right\} \rightarrow 1.$$

*For  $p = 0$ ,  $Q_{\text{hist}}$  is polynomial in  $n$  for fixed  $t$ .*

*Proof.* For fixed  $t$ ,  $\log \binom{n}{t} = O(\log n)$ , so  $L_{n,t} = O(\log n)$ . Since  $E/m \rightarrow p$  in probability,  $w_E/m = p + o_{\text{Pr}}(1)$ . For any  $q < 1/2$ ,

$$\sum_{i=0}^{\lfloor qm \rfloor} \binom{m}{i} \leq 2^{mh_2(q)}.$$

The prefactor  $2^t \binom{n}{t}$  has normalized logarithm  $o(1)$ , and continuity of  $h_2$  completes the proof.  $\square$

*Remark 1* (What the theorem does not prove). The theorem bounds history generation, not processor cycles, memory traffic, or the performance of unrestricted edit rates. It also does not prove optimality. A resource converse under a declared oracle model remains a target. For  $t$  growing with  $n$ , the deletion-subset factor may alter the exponent.

## 5 Independent strong exact baselines

The large speedups against exhaustive codeword ML are insufficient because the reference cost grows as  $2^k$ . Version 2 therefore compares the membership-only decoder with two code-specific exact searches that do not materialize the codebook.

### 5.1 Syndrome-trellis aggregate search

For a systematic linear code with parity-check matrix  $H$  and a fixed deletion alignment  $j$ , a codeword path selects  $x_i \in \{0, 1\}$  while updating the syndrome. Survivor coordinates incur mismatch cost  $1\{x_i \neq y_{\pi_j(i)}\}$  and the deleted coordinate has zero cost. A backward dynamic program computes the exact minimum remaining cost for every  $(i, s)$  state.  $A^*$  then emits codewords for alignment  $j$  in nondecreasing mismatch cost. The  $n$  alignment streams are merged, candidates are exactly aggregate-scored, and the same residual sum certificate is applied.

This is an independent code-specific baseline because it exploits the full parity-check trellis rather than issuing generic membership queries. Its preprocessing is charged per received word.

### 5.2 Code-aware aggregate prefix $A^*$

A second baseline searches candidate prefixes directly. Its state stores the prefix, all alignment mismatch counts, and systematic-code prefix feasibility. The priority key is an admissible upper bound obtained by assuming every unassigned survivor symbol can match perfectly. Exact terminal scores and a strict frontier certificate produce ML decoding.

The authoritative reference at each trial is the faster of the syndrome-trellis and code-aware prefix implementations. This avoids declaring victory over a deliberately weak baseline.

### 5.3 Specialized VT control

For a noiseless one-deletion channel and a VT code, direct decoding can enumerate the  $2n$  single-bit insertions, evaluate the VT checksum, aggregate repeated candidates, and select the largest multiplicity. FIBER–GRAND is not required to beat this specialized method; it must remain within a modest constant factor while retaining the generic interface.

## 6 Frozen computational protocol

No primary benchmark constructs a dictionary of all codewords. Linear membership is computed by syndrome; VT membership by checksum. Code construction and per-received-word baseline preprocessing are charged or explicitly recorded.

## 7 Decision contract

### 7.1 Hard failures

The field-defining programme stops immediately if any of the following occurs:

1. a false certificate or disagreement with exhaustive ML in a complete micro-world;
2. disagreement among independently certified membership-only, syndrome-trellis, and prefix  $A^*$  decisions;
3. violation of Theorem 1’s deterministic inequality or an observed history count exceeding (4);

Table 1: Standard version-2 gate.

| Component              | Frozen setting                                                                                   |
|------------------------|--------------------------------------------------------------------------------------------------|
| Primary channel        | One uniform deletion plus BSC, $p \in \{0.02, 0.05, 0.10\}$                                      |
| Codes                  | Random systematic linear and deterministic CRC-like linear families                              |
| Lengths/rate           | $n \in \{16, 20, 24, 28, 32\}$ , $R \approx 0.75$                                                |
| Membership-only method | History enumeration, syndrome membership, exact aggregate score, residual certificate            |
| Strong baselines       | Syndrome-trellis aggregate $A^*$ and code-aware prefix $A^*$ ; faster wall time is authoritative |
| Breadth                | Two deletions at $n \leq 24$ ; VT controls at $n \in \{15, 31, 63\}$                             |
| Theory audit           | $t \in \{1, 2\}$ , $n \leq 96$ , 200 binomial error samples per $(n, t, p)$                      |
| Correctness            | Exhaustive micro-worlds plus independent agreement of certified algorithms                       |
| Metrics                | Raw work components, wall time, completion, tails, membership fraction, theorem bound            |

4. hidden materialization of the primary codebook; or
5. the strongest code-specific baseline is more than five times faster in every required largest-length configuration.

## 7.2 Continuation threshold

Continuation to the final gate requires all of:

1. exact completion in every required primary configuration;
2. median membership-only wall time no more than three times the faster code-specific exact baseline for both code families at  $p = 0.02$  and  $p = 0.05$ ;
3. at least two required configurations in which the membership-only decoder is at least  $1.25\times$  faster;
4. exact VT agreement and median wall overhead no greater than  $5\times$ ;
5. at least 95% two-deletion completion with median membership fraction below 0.10;
6. decreasing finite-length overhead of the shell bound toward  $h_2(p)$ , with all deterministic inequalities valid.

These are research-investment thresholds, not universal constants.

## 8 Reference validation and interpretation

The release-standard reference run passed the revised gate. At  $n = 32$ , all required configurations had exact agreement. For  $p = 0.02$ , membership-only search was roughly five to six times faster than the faster code-specific exact baseline; for  $p = 0.05$ , it was roughly two to two-and-a-half times faster in median wall time. The 95th-percentile ratio was less favorable, demonstrating that latency tails remain a central issue. The VT implementation was within roughly  $1.3\text{--}1.9\times$  in median wall time. Two-deletion completion was exact in the frozen small sample, but its high-ambiguity tail at  $p = 0.05$  remained large.

These numbers are not publication claims until reproduced in the user’s environment and subjected to stronger external implementations. They establish that the v1 speedups were not solely an artifact of exhaustive codeword ML or a precomputed membership dictionary.

## 9 Final field-defining gate after a positive version-2 run

A positive version-2 result authorizes only four tightly scoped tasks.

### 9.1 N1: independent primary-source novelty adjudication

Produce a claim-by-claim matrix containing the closest primary source, exact overlap, surviving novelty, and verdict for:

1. the fixed-edit shell certificate and  $h_2(p)$  history-work exponent;
2. membership-first exact aggregate inverse search;
3. the search-inflation and history-duplication quantities;
4. code modularity relative to priority-first and synchronization trellis decoding;
5. the proposed bounded-ambiguity tractability hierarchy.

If the shell theorem or complete synthesis is already established, the field-defining claim stops or is renamed before further investment.

### 9.2 N2: independent proof review

An external reviewer must check every inequality, strictness/tie condition, shell count, asymptotic step, and oracle assumption. A proof defect that cannot be repaired without weakening the exponent statement is a STOP.

### 9.3 N3: additional specialized system baselines

At least one matched baseline from each relevant class must be implemented or obtained from authors:

- VT/shifted-VT or direct single-edit decoding;
- GC+ or Guess-and-Check;
- marker/watermark plus forward–backward/BCJR;
- a joint channel–code trellis or priority-first decoder;
- a modern insertion/deletion construction such as polar/guard-band where the operating regime matches.

No baseline may be denied channel parameters, soft information, code structure, or optimized implementation available to FIBER–GRAND.

### 9.4 N4: calibrated post-front-end impairment

Select one primary telecommunications mechanism rather than a generic deletion abstraction. The strongest candidates are:

1. variable-length demapping errors from nonuniform labeling;
2. bounded symbol slips after a timing/synchronization loop;
3. short-packet frame-boundary ambiguity;



4. a measured low-power receiver’s missed/duplicated symbol decisions.

The front end must output an alignment lattice or edit prior and independent held-out traces. Continuation requires a reproducible regime in which aggregate code-aware synchronization provides a material latency, energy, or reliability advantage over the best matched system baseline.

## 9.5 Final stop rule

The programme stops as a field-defining investment if any one of N1–N4 fails. Passing N1–N4 would establish a credible candidate programme, not guarantee retrospective field-defining impact.

## 10 Use-case boundary

The revised mechanism is aimed at short, high-rate blocks with a small edit budget and uncertain alignment. Plausible scenarios include timing slips, variable-length demapping, framing ambiguity, packet duplication/loss over short coded sequences, and DNA/storage as a secondary benchmark. It is not a general replacement for LDPC/polar decoders on ordinary memoryless channels, nor is it claimed efficient for a positive edit fraction or unrestricted ambiguous transducer.

## 11 Reproducibility requirements

Every release must preserve:

1. exact micro-world tests and zero false certificates;
2. per-trial history, candidate, membership, score, trellis, wall, and memory records;
3. explicit code-interface level and whether any codebook was materialized;
4. source commit, environment, seeds, checksums, and immutable decision thresholds;
5. mean and high-quantile latency, not only average speedup;
6. all negative regimes, including cases where the specialized decoder is superior.

## 12 Primary-source boundary

The final novelty audit must begin with the following established foundations:

- K. R. Duffy, J. Li, and M. Médard, “Capacity-achieving guessing random additive noise decoding,” *IEEE Trans. Inf. Theory*, 2019.
- R. G. Gallager, “Sequential decoding for binary channels with noise and synchronization errors,” MIT Lincoln Laboratory, 1961.
- Y. S. Han, C. R. P. Hartmann, and C.-C. Chen, “Efficient priority-first search maximum-likelihood soft-decision decoding of linear block codes,” *IEEE Trans. Inf. Theory*, 1993.
- M. C. Davey and D. J. C. MacKay, “Reliable communication over channels with insertions, deletions, and substitutions,” *IEEE Trans. Inf. Theory*, 2001.
- M. Mohri, “Weighted automata algorithms,” in *Handbook of Weighted Automata*, 2009.

- C. de la Higuera and J. Oncina, “The most probable string: an algorithmic study,” *Journal of Logic and Computation*, 2014.
- O. Sabary et al., work on ML and decoding error over one/two deletion channels, including fixed- $k$  deletion models.
- B. Haeupler and A. Shahrasbi, survey of synchronization strings and insertion/deletion codes, and contemporary VT, GC+, marker/watermark, and polar synchronization literature.
- E. Arıkan, “An inequality on guessing and its application to sequential decoding,” *IEEE Trans. Inf. Theory*, 1996, together with later conditional-guesswork refinements.

## 13 Conclusion

The first gate did not prove a new field, but unlike the two rejected programmes it produced a real exact computational mechanism across several edit channels and code families. The revised analysis identifies a potentially stronger foundational result: for a fixed number of deletions, certified aggregate history search inherits the BSC substitution-guessing exponent up to fixed-edit polynomial overhead. The version-2 package tests this theorem’s implementation and compares the membership-only decoder against exact code-specific searches without materializing the codebook.

A passed version-2 gate justifies one final, externally audited investment step. It does not justify hardware, unrestricted transducers, or a field-defining headline. Failure of the final novelty, proof, baseline, or physical-premise gate triggers an immediate stop.